

HEALTHCONNECT CLINIC

Machine Learning Problem Definition

AnalystLab Africa Experience Lab — Week 4

Data Science Track

Introduction

Missed healthcare appointments can create problems for both patients and healthcare providers.

When a patient does not attend a scheduled appointment, the appointment slot may go unused. This can affect clinic efficiency and make appointment planning more difficult.

HealthConnect Clinic wants to explore how data and artificial intelligence can help reduce missed appointments and improve patient support. The main project question is how the clinic can use data and AI to improve appointment attendance and the overall patient experience.

For the Data Science track, my role is to determine whether the available appointment data can support a machine learning model for predicting appointment no-shows.

Week 4 is focused on defining this problem and planning the approach. Model development will come later.

1. Business Problem

HealthConnect Clinic has patients who schedule appointments but do not attend them.

A no-show can leave an appointment slot unused. It can also make scheduling and resource planning more difficult.

The clinic therefore needs a better way to identify appointments that may have a higher risk of becoming no-shows.

Machine learning could potentially help with this problem.

Instead of treating every appointment in the same way, a future model could estimate the probability that a scheduled appointment will become a no-show.

This information could then support administrative actions such as targeted reminders or follow-up.

The model would support decision-making. It would not make medical decisions about patients.

2. Machine Learning Problem

The proposed machine learning task is a supervised binary classification problem.

The model will use information available before an appointment to predict whether that appointment is likely to result in:

0 — Attended

or

1 — No-Show

The main prediction question is:

Can information available before a scheduled appointment be used to predict whether the patient will attend or become a no-show?

This directly supports the Data Science responsibility defined for Week 4: assessing whether the available data can support a no-show prediction solution.

3. Dataset Overview

The HealthConnect Appointment Dataset contains fictional and anonymized appointment records.

The available information covers patient demographics, appointment details, booking information, previous appointment history, previous no-shows, reminders, distance to the clinic, estimated waiting time, and appointment outcomes.

The dataset contains:

5,000 appointment records	18 variables	1,696 unique patients
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The difference between the number of appointments and patients is important.

Some patients have more than one appointment in the dataset. This needs to be considered later when the data is divided into training and testing sets.

4. Initial Data Quality Assessment

The initial assessment showed that the dataset is generally well structured.

There were:

- 0 duplicate rows
- No major logical problems with the appointment dates
- No invalid historical no-show counts
- Consistent booking lead-time calculations

Three variables contained missing values:

Variable	Missing Values	Percentage
reminder_channel	1,366	27.32%
distance_to_clinic_km	90	1.80%
waiting_time_minutes	60	1.20%

The missing values should not all be treated in the same way.

5. Understanding the Missing Reminder Information

reminder_channel has the largest number of missing values.

At first, this could look like a serious data-quality problem.

However, further investigation showed that all 1,366 missing reminder channels occurred when reminder_sent = No.

This means the missing values have a logical explanation.

If no reminder was sent, there is no reminder channel to record.

For this reason, these values should not simply be replaced with the most common reminder channel.

They represent structural missingness and should be handled in a way that preserves their meaning.

The smaller amount of missing information in distance_to_clinic_km and waiting_time_minutes will require an appropriate treatment strategy during preprocessing.

6. Data Consistency Checks

Several logical checks were performed.

Previous appointment history

previous_no_shows should never be greater than previous_appointments.

No violations were found.

Invalid records: 0

Booking and appointment dates

A booking date should not occur after its scheduled appointment date.

No violations were found.

Invalid records: 0

Booking lead time

booking_lead_days was compared with the actual difference between booking_date and appointment_date.

All values matched.

Inconsistent records: 0

These results increase confidence in the overall quality of the dataset.

7. Proposed Target Variable

The original outcome variable is:

appointment_outcome

It contains three possible outcomes:

- Attended
- No-Show
- Cancelled

The distribution is:

Appointment Outcome	Records	Percentage
No-Show	2,423	48.46%
Attended	2,314	46.28%
Cancelled	263	5.26%

The main business problem is specifically about no-shows.

For this reason, I propose creating a new binary target:

is_no_show

Attended → 0

No-Show → 1

Cancelled appointments will initially be excluded.

8. Why Cancelled Appointments Will Be Excluded

A cancelled appointment is different from a no-show.

In a cancellation, the clinic is informed that the patient will not attend.

This may give the clinic an opportunity to reschedule the patient or reuse the appointment slot.

A no-show is different because the appointment remains scheduled but the patient does not attend.

Combining these two behaviours could make the prediction problem less clear.

For the first model, the goal will therefore be:

No-Show vs Attended

After removing cancellations, 4,737 appointments remain for the proposed binary modelling task.

The two classes are also close in size.

This means severe class imbalance is not expected to be a major problem in the initial model.

Cancellations could be investigated separately in a later stage.

9. Potential Input Features

The dataset contains several variables that could potentially help predict appointment attendance.

Patient Information

Age

Age may be related to appointment behaviour and is available before the appointment.

Gender

Gender may also be explored as a candidate feature. However, its use should be assessed carefully for fairness.

Appointment Information

Appointment Type

Different appointment types may have different attendance patterns.

Appointment Day

Attendance may vary depending on the day of the week.

Appointment Time

Morning, afternoon, and evening appointments may show different patterns.

Booking Information

Booking Lead Days

This represents the number of days between booking and the appointment.

A longer waiting period between booking and attendance may potentially affect the chance of a no-show.

Patient Appointment History

Previous Appointments

This provides information about the patient's previous interaction with the clinic.

Previous No-Shows

This may be particularly useful because previous attendance behaviour could help identify future appointment patterns.

Reminder Information

Reminder Sent

This shows whether a reminder was sent before the appointment.

Reminder Channel

This records how the reminder was delivered when one was sent.

Access and Operational Information

Distance to Clinic

Longer travel distance may affect how easy it is for a patient to attend.

Estimated Waiting Time

The Data Dictionary defines `waiting_time_minutes` as an estimated waiting time.

Since this information can potentially be known before the appointment, it can be considered as a candidate predictor.

10. Variables That Should Not Be Used Directly

Not every column should automatically become a model feature.

Appointment ID

`appointment_id` is a unique identifier.

It does not describe patient behaviour or appointment characteristics.

It should therefore be excluded from the model.

Patient ID

`patient_id` should also not be used as a normal predictive feature.

However, it is still important.

The same patient can appear across multiple appointment records. Patient ID can therefore help prevent the same patient's appointments from being improperly distributed across training and testing data.

Age Group

`age_group` is derived from `age`.

Keeping both may introduce unnecessary redundancy.

The continuous Age variable will initially be preferred because it preserves more information.

Raw Dates

`booking_date` and `appointment_date` should not automatically be passed into the model as raw dates.

Useful information can instead be extracted from them where needed.

The dataset already contains useful derived information such as `booking_lead_days` and `appointment_day`.

Appointment Outcome

`appointment_outcome` is the source of the target.

It must never be included as an input feature because this would cause direct target leakage.

11. Prediction Timing and Data Leakage

One of the most important decisions is defining when the prediction will happen.

The model should only use information that would realistically be available at the time a prediction is made.

For the initial approach, I propose making the prediction shortly before the scheduled appointment.

At this point, reminder information may already be available.

This allows `reminder_sent` and `reminder_channel` to be considered as candidate features.

If the clinic later decides that predictions should be made immediately when an appointment is booked, reminder information would no longer be available and should be excluded.

This prediction point must therefore remain clearly defined during model development.

12. Repeated Patients

The dataset contains 5,000 appointments but only 1,696 unique patients.

This means some patients appear several times.

A simple random train-test split could place one appointment from a patient in the training set and another appointment from the same patient in the test set.

This could make model evaluation less realistic.

A patient-level group split should therefore be considered during model development.

This would help ensure that the model is evaluated on patients it did not already see during training.

13. Initial Modelling Approach

The proposed modelling workflow is:



A simple Logistic Regression model could be used as the baseline because it is easy to interpret.

Other models could then be compared, such as:

- Decision Tree
- Random Forest

- Other suitable classification algorithms

The final model should be selected based on validation performance and practical usefulness, not simply because it is more complex.

14. Proposed Evaluation Metrics

Accuracy alone will not be enough.

The main purpose of the model is to identify appointments at risk of becoming no-shows.

Several metrics should therefore be considered.

Recall

Recall will show how many actual no-shows were correctly identified.

This is important because missing many high-risk appointments would reduce the usefulness of the system.

Precision

Precision will show how many appointments predicted as no-shows actually became no-shows.

Low precision could lead to too many unnecessary interventions.

F1-Score

F1-score provides a balance between precision and recall.

ROC-AUC

ROC-AUC can help measure how well the model separates attended appointments from no-shows across different thresholds.

The final decision threshold should eventually depend on how the clinic plans to use the predictions.

15. Assumptions

The initial approach makes several assumptions.

It assumes that the historical appointment information is accurate.

It assumes that candidate predictors are available before the prediction is made.

It also assumes that past appointment behaviour may provide useful information about future attendance.

The proposed prediction time is shortly before the scheduled appointment, after reminder information may be available.

These assumptions should be reviewed during later stages of the project.

16. Limitations

The dataset is fictional and anonymized.

Results from this project should therefore not be treated as evidence that the same patterns would occur in a real healthcare population.

Some variables contain missing information.

The dataset also contains repeated appointments from the same patients.

In addition, the available variables may not capture every factor that influences whether a patient attends an appointment.

17. Risks and Ethical Considerations

A no-show prediction model should support patients rather than punish them.

A patient may miss an appointment because of transport problems, work responsibilities, financial challenges, family responsibilities, or other circumstances that are not fully represented in the dataset.

Demographic variables such as age and gender should therefore be evaluated carefully for potential bias.

Predictions should be used to support actions such as reminders or administrative assistance.

They should not be used to deny care or make clinical decisions.

18. Overall Data Suitability

Based on the initial assessment, the HealthConnect Appointment Dataset appears suitable for developing an initial no-show prediction model.

It has several strengths:

- A clearly defined appointment outcome
- A reasonable number of records
- Relevant pre-appointment variables
- Patient appointment history
- Reminder information
- Limited missingness in most features
- No duplicate records
- Consistent date and historical information
- A nearly balanced target after cancellations are excluded

The main issues that require careful handling are missing values, repeated patients, prediction timing, feature availability, cancellations, and potential bias.

These issues are manageable and do not prevent the project from moving to the next stage.

19. Proposed Focus for the Next Phase

The next stage should move from problem definition into deeper analysis and solution development.

The main priorities should be:

- Explore the relationship between candidate features and no-show behaviour.
- Handle missing values appropriately.
- Prepare categorical and numerical variables.
- Investigate useful feature engineering opportunities.
- Develop a leakage-safe train-test strategy.
- Build an interpretable baseline model.
- Compare additional classification models.
- Evaluate performance using suitable metrics.
- Examine feature importance and model behaviour.

Conclusion

The goal of this project is not simply to predict who will miss an appointment.

The real goal is to provide HealthConnect Clinic with useful information that could help improve appointment attendance and make better use of available appointment slots.

The initial assessment shows that the available data can support this goal.

A binary classification model will be developed to predict No-Show vs Attended appointments. Cancelled appointments will initially be treated separately.

The dataset contains useful information about demographics, appointment characteristics, booking behaviour, previous attendance, reminders, distance, and estimated waiting time.

The next stage will test whether these variables can produce reliable and useful predictions.

The final model should remain interpretable, carefully validated, and focused on supporting better patient engagement rather than replacing human decision-making.